San Jose Formation | Research Source Brief

Cuba, New Mexico • Peace Engineering water-resources project

Source and main finding

Blandford et al. (2025), "San Jose Formation Hydrostratigraphy and Groundwater Conditions in the Vicinity of Lindrith, New Mexico," New Mexico Geological Society Guidebook 75, pp. 235–246. The study analyzes approximately 840 km^2 near Lindrith/Apache Ranch, using >350 geophysical logs and 85 water-well drilling logs. It identifies four San Jose hydrostratigraphic units, maps the thickness of Unit 4 (primarily the Cuba Mesa Member), and develops regional groundwater-level surfaces. The proposed deep irrigation supply lies about 300–425 m below ground.

Research-question relevance

Question	Support	How to use this publication
Q4: Ojo Alamo vs. San Jose	STRONG	Compare San Jose/Cuba Mesa extent, thickness, layering and hydraulic-head patterns with the separate Ojo Alamo USGS study; note different study areas.
Q2: Well depths and aquifer units	STRONG	Use the hydrostratigraphic interpretation and drilling-log evidence to explain which units wells penetrate; this is not a municipal well inventory.
Q8: Monitoring coverage and gaps	MODERATE	Use mapped well/log locations and differences between shallow and deep observations to discuss where evidence is limited.
Q9: Water-stress dashboard	CONTEXTUAL	Add aquifer-unit and thickness context to a map; the publication alone cannot quantify current water stress.
Q1 / Q3: Trends / Cuba depletion	LIMITED	Water-level surfaces are not a time series proving long-term decline or depletion of Cuba's municipal supply.

Is it relevant to Cuba's water sources?

YES — regionally and geologically. The Cuba Mesa Member is a relevant candidate groundwater-bearing unit for investigating aquifers around Cuba. NOT VERIFIED — the study is centered near Lindrith, not Cuba's municipal wells; it does not demonstrate that Cuba pumps from this unit or that its drinking-water aquifer is being depleted. Well completion records are required to establish a direct connection.

Suggested report figures and next steps

Ask the plotting teammate to extract or reproduce, with attribution, (1) the Cuba Mesa/Unit 4 aquifer-thickness map, (2) the hydrostratigraphic cross-section or unit map, and (3) groundwater-level surface maps. Overlay Cuba and Lindrith only if map coordinates and scale permit. Pair this source with the Ojo Alamo Sandstone USGS publication for Q4; request Cuba municipal well logs before making claims about its actual source or depletion.

Source: New Mexico Geological Society, 2025. https://nmgs.nmt.edu/publications/guidebooks/details.cfml?ID=134492 | DOI: 10.56577/FFC-75.235